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Hyperparameter-Tuned Lightweight CNN-SVM Hybrid Model for Efficient Multi-Class Lung Cancer Classification

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***Abstract*— Lung cancer is a leading cause of cancer death worldwide; therefore, it is very crucial to have an early and accurate diagnosis for effective treatment. Existing research primarily focuses on binary classification (benign vs. malignant), often overlooking the need to distinguish normal lung from benign lung. Additionally, while deep learning models like ResNet and DenseNet perform very well in classification tasks, their computational complexity poses challenges in real-world applications. To address these issues, our study proposes a lightweight cascade model combining a custom CNN (Convolutional Neural Network) as a feature extractor with SVM (Support Vector Machine) as a benign, malignant, and normal classifier. Moreover, the Hyperband algorithm is employed for hyperparameter tuning, and Xavier initialization is used for initializing the model's weights and biases. Four different datasets are merged, and several image preprocessing techniques are used to focus on the Region of Interest (ROI). Furthermore, the data augmentation techniques are used to resolve the class imbalance issue. The experimental results demonstrate that the proposed CNN-SVM hybrid model outperforms traditional single-model and complex hybrid model approaches, achieving a superior accuracy of 0.966, a precision of 0.97, a recall of 0.97, and an F1 score of 0.97, while maintaining computational efficiency. The final model's performance is evaluated through comparative evaluation, confusion matrices, and loss curves. This study provides a promising direction for developing efficient, high-accuracy lung cancer classification models for clinical applications.**

***Index Terms*—ResNet, DenseNet, CNN, SVM, Hyperband, Xavier Initializer.**

I. Introduction

Cancer is a major global health problem affecting millions [1]. According to the International Agency for Research on Cancer (IARC), a specialized agency of the World Health Organization (WHO) dedicated to cancer research, lung cancer is the leading cause of cancer-related deaths, as reported in the Globocan 2022 report [2]. A huge number of patients are encountered by radiologists on a regular basis nowadays. Manual inspection of lung CT scans can be very time-consuming. Therefore, computer-aided diagnosis (CAD) systems are essential for modern radiologists to reduce their daily workload. Over the years, numerous researchers have worked extensively to enhance the accuracy and timeliness of lung cancer detection. However, most studies primarily focus on classifying lung conditions into benign or malignant categories. Benign tumors, while capable of growing significantly, do not invade adjacent tissues or spread to other parts of the body. They typically exhibit well-defined borders and a smooth shape [3]. In contrast, a healthy lung contains neither benign nor malignant tissue. Therefore, distinguishing among benign, malignant, and normal lung conditions is essential for developing a robust and sophisticated CAD system.

Since lung nodules are very small in nature [4], researchers tend to use state-of-the-art models such as ResNet, DenseNet, etc., for capturing those small features. However, those models are computationally very complex, resulting in more time to diagnose. This is contradictory to the purpose of CAD systems. Complexity also causes those models to suffer from the gradient vanishing problem. Consequently, this reduces the model's performance.

Moreover, researchers are not effectively leveraging the strength of image processing, whereas it can be used to enhance the image features. Image enhancement and morphological techniques are very effective [5] for processing medical images before feeding them to the model.

This study proposes a lightweight cascade model for lung classification, expanding beyond the conventional binary classification to categorize lung conditions into three distinct classes utilizing image processing techniques, thereby enhancing diagnostic precision and clinical decision-making.

The rest of this paper is outlined as follows. In section II, we discuss the recent trend and literature review, while in section III, the methodology is described. The experiment and result are presented in section IV. Lastly, section V concludes the paper and addresses future works.

II. Survey of the Literature

Lung cancer is a life-threatening disease, and manual investigation of lung CT (computed tomography) scans is time-consuming, making it a critical area of research. Over the years, researchers have explored various approaches, including lung nodule detection, segmentation, and classification. They have explored various approaches, ranging from classical machine learning models like SVMs to deep learning architectures such as CNNs and pretrained models like ResNet and DenseNet. Furthermore, the cascading approach, where multiple models are used in succession, has been investigated to improve classification accuracy. However, this technique

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comes with a trade-off—while it enhances performance, it also increases computational cost.

Makaju et al. [6] employed SVM for binary lung cancer classification (benign vs. malignant) using the Lung Image Database Consortium-Image Database Resource Initiative (LIDC-IDRI) dataset [7]. They use an image smoothing filter that may remove the lung nodule or remove the sharpness of the nodule. Although they got high performance with a simple machine learning model, in terms of image deep learning outperform SVM. Nanglia et al. [8] introduced a hybrid model that integrates SVM with a Feed-Forward Back Propagation Neural Network for the ELCAP lung image dataset. Their approach effectively reduced computational overhead, achieving 98.17% precision, 98.08% accuracy, 96.5% recall, and 97% F-measure over 500 CT images. The main limitation of this work is that they have used a very small dataset. They classify them into two classes, benign and malignant, and don't care about normal lungs.

Asuntha and Srinivasan [9] proposed a lightweight CNN called FPSOCNN (Fuzzy Particle Swarm Optimization Convolutional Neural Network) to reduce computational complexity. The FPSO algorithm was used for optimal feature selection. Using the LIDC/IDRI dataset, they classified lung cancer as benign or malignant. Whereas the LIDC/IDRI dataset has some scans that does not contain any nodules. They don't deal with those normal lungs. Chiet et al. [10] investigated several pretrained CNN models, including ResNet-50, VGG-16, Xception, and MobileNet, to classify lung cancer using the LIDC/IDRI dataset. Their models achieved state-of-the-art accuracy, ranging from 78% to 86%, in distinguishing cancerous vs. non-cancerous lung images; they also don't incorporate normal lung.

Lanjewar et al. [11] applied a DenseNet classifier to a chest CT-scan dataset for multi-class lung cancer classification, identifying adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal cells. Feature selection techniques were used to extract the most relevant features from DenseNet201, which were then applied to multiple machine learning classifiers. They have used a very complex model; as a result, their model suffers from overfitting. Whereas Naseer et al. [12] developed LungNet-SVM, a modified AlexNet architecture combined with an SVM classifier for binary classification (benign vs. malignant). Their approach achieved 97.64% accuracy, 96.37% sensitivity, and 99.08% specificity. Normal lungs are not included, even though they used a relatively complex model.

Most of the existing studies focus on binary classification (benign vs. malignant). However, it is crucial to introduce a third class—normal cases, which distinguishes healthy individuals from patients with lung abnormalities. Additionally, multiple studies have demonstrated that combining models in a cascading fashion improves classification accuracy, yet they undergo the gradient vanishing problem. However, increased computational complexity remains a significant challenge.

This research aims to design a lightweight hybrid model that not only reduces computational cost but also enhances classification performance by distinguishing normal, benign, and malignant lung conditions.

III. Materials and Methods

Our research work goes through several pipelines. Firstly, we prepare our candidate dataset by merging four different datasets. Then we conducted several image preprocessing techniques to increase the necessary features. Finally, we designed our proposed model architecture and trained it on preprocessed data. The whole research workflow has been shown in Figure 1.

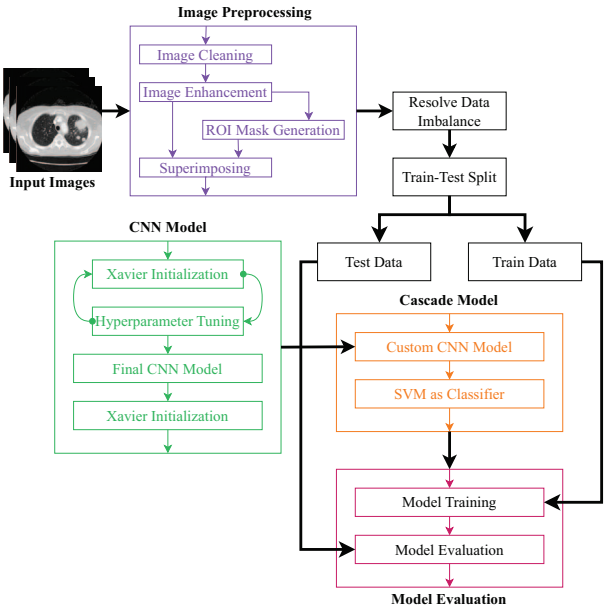


Fig. 1: Proposed research pipeline showing overall workflow of our research.

A. Dataset

This study utilizes four different publicly available datasets. Vertically merge them to enhance generalization, and a horizontal merge is used as well to achieve our target classes. The descriptions of those datasets are as follows:

- Dataset 1: Lung Cancer Detection—LIDC-IDRI Subset [13] [7], was compiled by Bushara, an assistant professor at KMEA Engineering College, India, and is derived from the LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) dataset. It compresses 2066 2D images of 512×512 size.
- Dataset 2: Chest CT-Scan Images Dataset [14], was developed by Mohammad Hany from Misr University for Science and Technology, Egypt. This dataset was designed for lung cancer detection. It has a total of 1000 2D images of different sizes.
- Dataset 3: The IQ-OTH/NCCD Lung Cancer Dataset [15] was collected from the Iraq-Oncology Teaching Hospital and the National Center for Cancer Diseases

(IQ-OTH/NCCD). This dataset was gathered over three months in the fall of 2019 from specialized hospitals. It collects 1097 2D images of 512x512 size.

- Dataset 4: CT Scan Images for Lung Cancer [16], was created by Dishant Rathi, an Indian researcher, for classifying and diagnosing lung cancer using deep learning techniques. It contains 2077 2D images of various sizes.

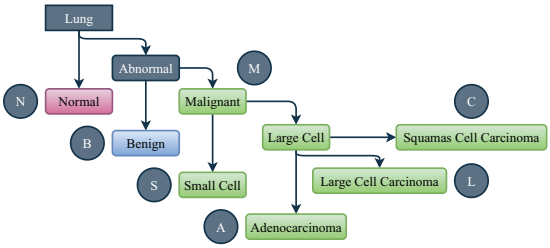


Fig. 2: Illustrated lung cancer classification proposed by WHO.

These datasets differ in various aspects, such as the number of samples, number of classes, image dimensions, and other characteristics. To ensure consistency and enhance model performance, image preprocessing techniques were applied before feeding the data into the proposed model. Each of the four datasets contains a varying number of classes. In this study, the datasets were merged based on the lung classification system proposed by WHO (World Health Organization) (illustrated in Figure 2), consolidating lung conditions into three primary categories: normal (N'), benign (B'), and malignant (M'). However, the WHO proposed a lung cancer classifier with six categories: normal (N), benign (B), small cell lung carcinoma (S), adenocarcinoma (A), large cell carcinoma (L), and squamous cell carcinoma (C). Whereas, datasets contain different number of classes among them, such as normal, benign, adenocarcinoma, large cell carcinoma, and squamous cell carcinoma. Nevertheless, those datasets contain an extra class named malignant (M) that represents any malignant categories of cancer illustrated in Figure 2. The summaries of those four datasets are presented in Table I. We merge them horizontally and vertically to prepare our proposed dataset.

TABLE I: Summary of our four candidate datasets.

Dataset	Different Classes					
	B	M	A	L	C	N
1	1015	1051	N/A	N/A	N/A	N/A
2	N/A	N/A	338	187	260	215
3	120	561	N/A	N/A	N/A	416
4	100	561	338	187	260	631

In the horizontal merging process, we merge all malignant (M), all adenocarcinoma (A), all large cell carcinoma (L), and all squamous cell carcinoma (C) images into the final malignant (M') class. Normal (N') and benign (B') classes are left as they are. If we classify lung cancer in detail into the six classes proposed by the WHO, this will be very effective for lung diagnosis. However, the reason behind choosing three classes instead of six is that the datasets do not contain small cell cancer types. Also, datasets 1 and 3 contain the malignant class that is not specific. Alternatively, in the vertical merging process, we combine all images from corresponding classes. In Table II, we depict our final dataset. Several preprocessing and data augmentation steps have been conducted on the final dataset. One sample image from each classes of the final dataset shown in Figure 3.

TABLE II: The description of our final dataset after merging the four candidate datasets.

Final Classes			Total Images	Image Size
B'	M'	N'		
1235	3743	1262	6240	Various

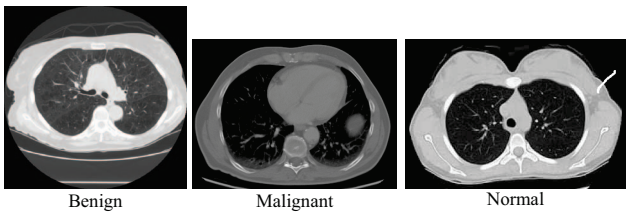


Fig. 3: Showing one sample per class from our final dataset.

B. Data Preprocessing

Since the final dataset is compiled from four different sources, it may contain redundant images. Eliminating these duplicates helps reduce computational costs during training by preventing the model from learning from the same image multiple times. To identify and remove redundant images, a hash function is employed. This function generates a unique hash value for each image. If two images produce the same hash value, they are considered identical, and one of them is removed.

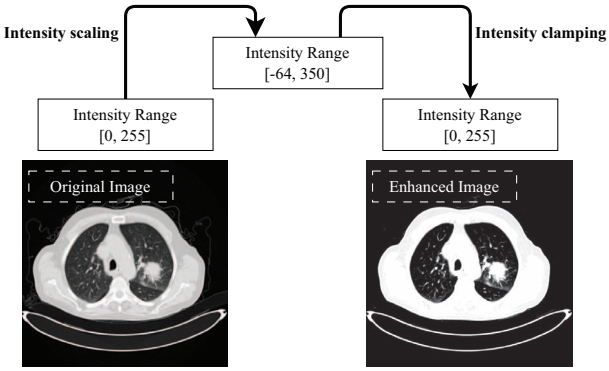


Fig. 4: Showing the effect of the enhancement process visually.

In grayscale images, each pixel holds an intensity value ranging from 0 to 255. To enhance image contrast, the intensity values are first extended to a range of -50 to 305 using

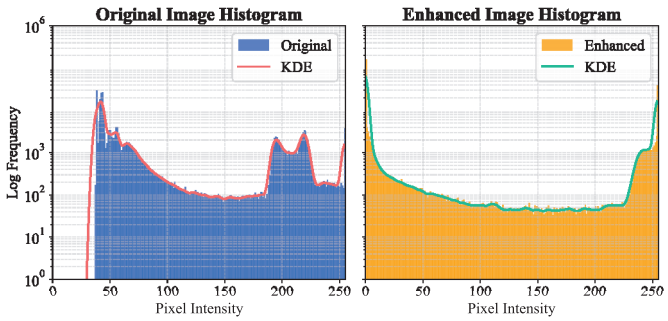


Fig. 5: The histogram of original and enhanced images. Showing the effectiveness of our enhancement process.

min-max normalization. This will make sure the high-contrast lung nodule. After this transformation, the values are clamped between 0 and 255 to ensure valid pixel intensity levels. This process enhances image contrast by making the darker regions appear even darker and the lighter regions appear brighter, thereby improving the visibility of important features for analysis. The image enhancement process is illustrated in Figure 4, and the effect of the enhancement process with respect to the histogram is shown in Figure 5.

To ensure consistency in image dimensions, all images are resized to a uniform shape. A manual inspection of the dataset revealed that while the largest image size is 512×512 pixels, many images have smaller dimensions. To preserve the aspect ratio, the resizing process (depicted in Figure 6) is performed in two steps. First, scaling—each image is proportionally scaled until either its height or width reaches 512 pixels. Second, zero padding—If necessary, zero padding is applied to extend the image to a final size of 512×512 pixels without distorting its original proportions. This process ensures the nodule's presence, and it preserves the nodule's actual shape.

C. Mask generation

Lung CT images are cross-sectional images of the human body that include various organs alongside the lungs. For this research, the lung region is the area of primary interest. To ensure that the model focuses solely on the lung, it is essential to isolate this region. Thus, a lung-segmented image is required for the proposed work. Image processing techniques such as erosion and dilation are employed to segment the region of interest, and this process involves several steps to achieve the final goal (illustrated in Figure 7). First, we negate the image (A). In a CT image, the lung appears darker compared to other organs. To make the lung portion more prominent, we negate the original CT image, converting the lung region to white. This negation ensures that only the lung area, with its corresponding intensity values, influences the model, facilitating a more focused learning process. Then, thresholding is applied (B) to convert the grayscale image into a binary image. This process requires a threshold value, which is computed using the K-means algorithm.

The K-means algorithm divides the intensity values into two clusters, and the mean of the cluster centers is used as

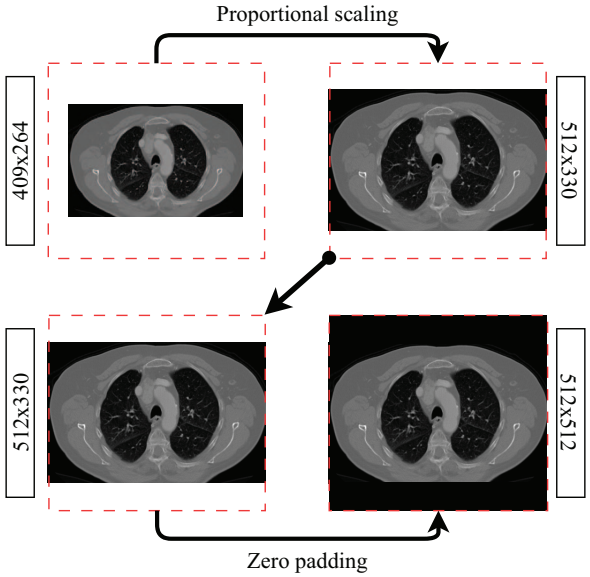


Fig. 6: The resizing process makes all images to 512x512.

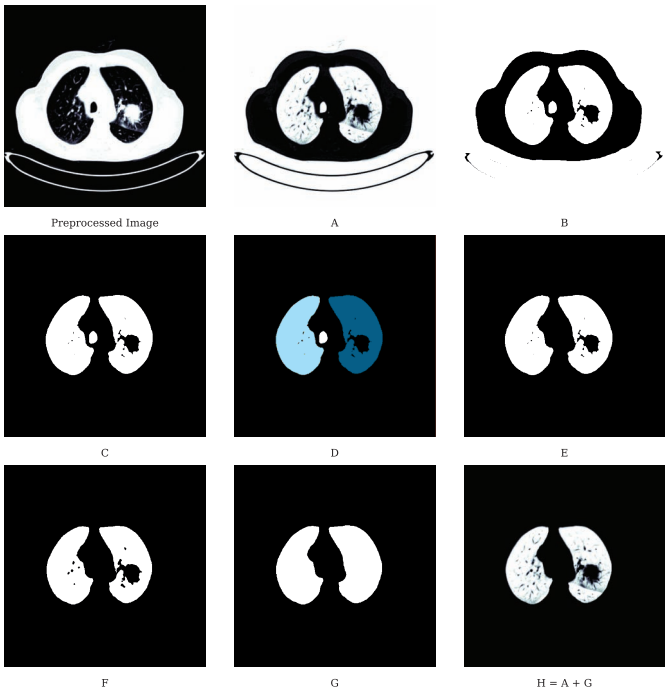


Fig. 7: Generating a 512x512 mask for each image using morphological operation.

the threshold value for segmentation. After that, the border clearing step is introduced (C), which involves clearing any connected objects near the lung region. After clearing, a region within the CT image remains, including the two lung lobes. In the next phase, each remaining region is labeled with a unique identifier (D). Each labeled region has properties such as its area, which helps distinguish different structures in the image. Then, all regions, except for the two largest ones, are removed (E). This ensures that only the two lung lobes remain in the image, isolating the desired region for further analysis. Next, lung nodules are separated from other structures, such as blood vessels, removing any sharp edges or lines that might interfere with the model's learning process (F). After that, the two largest regions, corresponding to the lung lobes, are cleaned to refine the final segmentation mask (G). This step ensures that the final mask accurately represents the lung region. Finally, the refined mask is superimposed on the negated image obtained in the first phase (H = A + G). This superimposition guarantees that only the relevant lung tissue is considered by the model, preventing irrelevant pixel intensities from affecting the model parameters.

TABLE III: Hyperparameter tuning result by using hyperband techniques.

Hyperparameter	Value
Number of layers	3
Nodes per layer	63, 64, 64
Batch size	21
Learning rate	0.0036457178424732185
Dropout rate	0.14052199966390633

D. Data balancing and splitting

Data augmentation is used to balance our dataset. As we mentioned earlier in the preprocessing section, redundant data has been removed. After removing redundant data, our dataset contains 1105 benign samples, 2277 malignant samples, and 470 normal samples. This exhibits a class imbalance issue. To mitigate this issue, image augmentation techniques have been applied to generate additional samples while preserving the original data distribution. Several augmentation techniques, such as flipping (horizontal and vertical), random rotation (15 degrees), brightness adjustment, and contrast adjustment, have been used, resulting in 1200 benign samples, 2277 malignant samples, and 1200 normal samples. Augmentation can't resolve the total imbalance problem. To overcome these difficulties, we have used weighted random sampling techniques to completely resolve the imbalance issue. The resultant dataset is split into 80% and 20% for training and testing, respectively.

E. Model architecture

This research employs a cascade model for lung classification, integrating a CNN and an SVM in a sequential manner. In this framework, the CNN functions as a feature extractor, while the SVM serves as the classifier. As outlined in the proposed methodology, this study employs hyperparameter tuning to optimize the performance of the lightweight custom CNN model. The Hyperband algorithm is utilized to efficiently search for the best set of hyperparameters. The summarized hyperparameters are listed in Table III. Additionally, Xavier initialization is applied to initialize the network parameters efficiently, ensuring better convergence. The proposed model architecture based on the resultant hyperparameter is shown in Figure 8. Our proposed model has a total of three convolutional layers with 64 nodes, along with maxpool layers. We have used ReLU as an activation function in the convolutional layers. The output of the last convolutional layer has been flattened into 1D features. Then, two dense layers associated with two dropout layers have been used to select important features from the resultant flattened features. The selected feature has been used for classification. We have used a linear kernel-based SVM for classifying lung cancer by using the selected features.

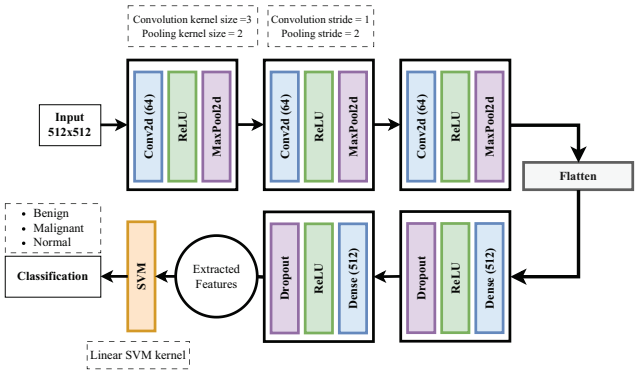


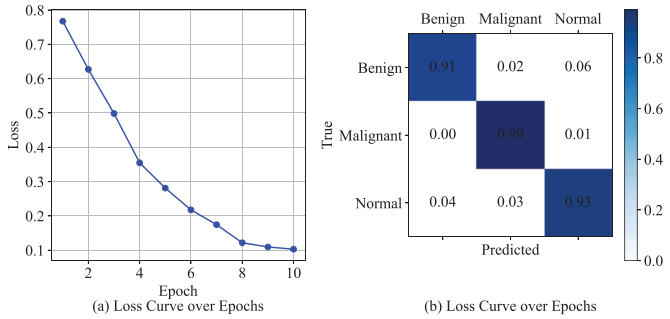
Fig. 8: The detailed architecture of our proposed model.

IV. Result Analysis

In classification tasks, the softmax head in the CNN model does not always provide sufficient performance. If we replace the softmax with a classical machine learning classification model like SVM or Random Forest (RF), then it can beat it.

We have already observed this trend in the literature review. However, in most cases, researchers used complex pretrained models. Due to the complexity, those models are suffering from the gradient vanishing problem. Our custom hybrid model overcame this issue and gave better performance than other approaches. The evaluation metrics have been shown in Figure 9.

We have trained and evaluated those models in our proposed dataset for comparison, like we have evaluated DenseNet and ResNet with a softmax head; further, we have trained them along with the SVM classifier head. Finally, our custom fine-tuned CNN feature extractor along with the SVM classification head. Our model achieved accuracy, precision, recall, and an F1 score of 0.9662, 0.97, 0.97, and 0.97, respectively. A comparative study between others' models and our proposed model is depicted in Table IV.



(a) Loss Curve over Epochs (b) Loss Curve over Epochs

Fig. 9: Training loss curve and confusion matrix for evaluation of our proposed model.

TABLE IV: Show the comparative result between others' models and our proposed model by using our final dataset.

Model	Accuracy	Precision	Recall	F1-score
ResNet50	0.9093	0.91	0.91	0.91
DenseNet121	0.8665	0.88	0.87	0.86
ResNet50 + SVM	0.9253	0.93	0.93	0.93
DenseNet + SVM	0.9235	0.93	0.92	0.92
Proposed model	0.9662*	0.97*	0.97*	0.97*

V. Conclusion and Future Works

The strengths of classical machine learning and deep learning have been utilized to create an efficient as well as lightweight hybrid approach for lung cancer classification. A typical lung classification task classifies lungs as benign or malignant. Our proposed approach classifies lungs as benign, malignant, or normal by leveraging four different public datasets. Hyperband tuning and Xavier initializer are used for efficient parameter and hyperparameter selection. By leveraging the hyperband tuning method, we get a lightweight hyperparameter for our proposed model. This hybridization demonstrates higher values of average precision, recall, and F1 score—those are 97%, 97% and 97%, respectively, for the lung cancer classification over 4677 samples of CT images. However, there are several future aspects of this research work, including extending our CNN model for 3D CT image datasets like the LIDC-IDRI dataset and adapting the SVM to increase the number of classes that can classify each category proposed by the WHO.

References

[1] Bray F. Global cancer statistics 2020: Globocan estimates of incidence and mortality worldwide for 36 cancers in 185 countries - sung - 2021 - ca: A cancer journal for clinicians - wiley online library. [Online; accessed 2025-05-01].

[2] Bray F. The globocan 2022 cancer estimates: Data sources, methods, and a snapshot of the cancer burden worldwide - filho - 2025 - international journal of cancer - wiley online library. [Online; accessed 2025-05-01].

[3] Differentiation of benign and malignant pulmonary nodules by growth rate — radiology. [Online; accessed 2025-05-01].

[4] Lung nodules: size still matters — european respiratory society, 12 2017. [Online; accessed 2025-05-01].

[5] Hamid Hassanpour, Najmeh Samadiani, and S.M. Mahdi Salehi. Using morphological transforms to enhance the contrast of medical images. *The Egyptian Journal of Radiology and Nuclear Medicine*, 46(2):481–489, 2015.

[6] Suren Makaju, P W C Prasad, Abeer Alsadoon, A K Singh, and A Elchouemi. Lung cancer detection using CT scan images. *Procedia Comput. Sci.*, 125:107–114, 2018.

[7] R. M. Nishikawa. The lung image database consortium (lidc) and image database resource initiative (idri): A completed reference database of lung nodules on ct scans - armato - 2011 - medical physics - wiley online library. [Online; accessed 2025-04-30].

[8] Pankaj Nanglia, Sumit Kumar, Aparna N Mahajan, Paramjit Singh, and Davinder Rathee. A hybrid algorithm for lung cancer classification using SVM and neural networks. *ICT Express*, 7(3):335–341, September 2021.

[9] A Asuntha and Andy Srinivasan. Deep learning for lung cancer detection and classification. *Multimed. Tools Appl.*, 79(11-12):7731–7762, March 2020.

[10] Chee Chiet Chai, Wee How Khoh, Ying Han Pang, and Hui Yen Yap. A lung cancer detection with pre-trained CNN models. *Journal of Informatics and Web Engineering*, 3(1):41–54, February 2024.

[11] Madhusudan G Lanjewar, Kamini G Panchbhai, and Panem Charanarur. Lung cancer detection from CT scans using modified DenseNet with feature selection methods and ML classifiers. *Expert Syst. Appl.*, 224(119961):119961, August 2023.

[12] Iftikhar Naseer, Tehreem Masood, Sheeraz Akram, Arfan Jaffar, Muhammad Rashid, and Muhammad Amjad Iqbal. Lung cancer detection using modified alexnet architecture and support vector machine. *Computers, Materials & Continua*, 74(1), 2023.

[13] Lung cancer detection_lidc-idri subset. [Online; accessed 2025-05-01].

[14] Performance comparison of vision transformer- and cnn-based image classification using cross entropy: A preliminary application to lung cancer discrimination from ct images. [Online; accessed 2025-05-01].

[15] The iq-oth/nccd lung cancer dataset. [Online; accessed 2025-05-01].

[16] Ct scan images for lung cancer. [Online; accessed 2025-05-01].